Python Programming - 2301CS404

Lab - 3

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01) WAP to check whether the given number is positive or negative.

In [4]:
```
number = int(input("Enter a Number : "))
if number > 0:
    print(f"{number} is positive")
elif number== 0:
    print(f"{number} is zero")
else:
    print(f"{number} is negative")
```

```
-5 is negative
```

02) WAP to check whether the given number is odd or even.

In [7]:
```
number = int(input("Enter a Number : "))
if number%2== 0:
    print(f"{number} is even")
else:
    print(f"{number} is odd")
```

```
5556 is even
```

03) WAP to find out largest number from given two numbers using simple if and ternary operator.

In [16]:
```
num1 , num2 =input("Enter two number with space :").split(" ")
num1=int(num1)
num2=int(num2)
print("using simple if operator:")
if num1 > num2:
    print(f"{num1} is greater than {num2}")
elif num1 == num2 :
    print(f"{num1} is equal to {num2}")
else:
    print(f"{num2} is greater than {num1}")

print("using ternary operator:")

print(f"{num1} is greater than {num2}") if num1 > num2 else print(f"{num1} is equal to {num2}") if num1 == num2 else print(f"{
```

```
using simple if operator:
5 is greater than 2
using ternary operator:
5 is greater than 2
```

04) WAP to find out largest number from given three numbers.

In [17]:
```
num1 , num2 , num3 =input("Enter two number with space :").split(" ")
num1=int(num1)
num2=int(num2)
num3=int(num3)
print("using simple if operator:")
if num1 > num2 and num1 > num3:
    print(f"{num1} is greater")
elif num2 > num3 :
    print(f"{num2} is greater")
else:
    print(f"{num3} is greater")

print("using ternary operator:")
```

```
print(f"{num1} is greater") if num1 > num2 and num1 > num3 else print(f"{num2} is greater") if num2 > num3 else print(f"{num3}
```

```
using simple if operator:
9 is greater
using ternary operator:
9 is greater
```

05) WAP to check whether the given year is leap year or not.

[If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year]

In [19]:
```
year = int(input("Enter a Year :"))
if (year%400==0) or (year%4==0 and year%100!=0):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")
```

```
2025 is not a leap year.
```

06) WAP in python to display the name of the day according to the number given by the user.

In [20]:
```
number = int(input("Enter a number for week day Name :"))
match number:
    case 1:
        print("Monday")
    case 2:
        print("Tuesday")
    case 3:
        print("Wednesday")
    case 4:
        print("Thursday")
    case 5:
        print("Friday")
    case 6:
        print("Saturday")
    case 7:
        print("Sunday")
```

```
    case _:
        print("Invalid Number")
```

```
Friday
```

07) WAP to implement simple calculator which performs (add,sub,mul,div) of two no. based on user input.

In [22]:
```
num1 , num2 =input("Enter two number with space :").split(" ")
operator = input("Enter an Operator(+,-,*,/)")
num1=int(num1)
num2=int(num2)
match operator:
    case "+":
        print(f"Addition of {num1} and {num2} = {num1+num2}")
    case "-":
        print(f"Subtraction of {num1} and {num2} = {num1-num2}")
    case "*":
        print(f"Multiplication of {num1} and {num2} = {num1*num2}")
    case "/":
        print(f"Division of {num1} and {num2} = {num1/num2}")
    case _:
        print("Invalid Operator")
```

```
Division of 5 and 2 = 2.5
```

08) WAP to read marks of five subjects. Calculate percentage and print class accordingly.

Fail below 35

Pass Class between 35 to 45

Second Class

between 45 to 60

First Class between 60 to 70

Distinction if more than 70

In [23]:
```
m1 , m2 , m3 , m4 , m5 = input("Enter marks of 5 subject with space :").split(" ")
m1=float(m1)
m2=float(m2)
m3=float(m3)
m4=float(m4)
m5=float(m5)
percentage = float( ( (m1+m2+m3+m4+m5) /500 ) *100 )
if percentage < 35:
    print(f"Percentage : {percentage} , Fail")
elif percentage >= 35 and percentage < 45 :
    print(f"Percentage : {percentage} , Pass")
elif percentage >= 45 and percentage < 60 :
    print(f"Percentage : {percentage} , Second")
elif percentage >= 60 and percentage < 70:
    print(f"Percentage : {percentage} , First Class")
else :
    print(f"Percentage : {percentage} , Distinction")
```

```
Percentage : 72.61999999999999 , Distinction
```

09) WAP to find the second largest number among three user input numbers.

In [24]:
```
num1 , num2 , num3 =input("Enter two number with space :").split(" ")
num1=int(num1)
num2=int(num2)
num3=int(num3)
if (num1 < num2 and num1 > num3) or (num1 > num2 and num1 < num3):
    print(f"{num1} is second largest")
elif (num2 < num1 and num2 > num3) or (num2 > num1 and num2 < num3):
    print(f"{num2} is second largest")
else:
    print(f"{num3} is second largest")
```

```
6 is second largest
```

10) WAP to calculate electricity bill based on following criteria. Which takes the unit from the user.

a. First 1 to 50 units – Rs. 2.60/unit

b. Next 50 to 100 units – Rs. 3.25/unit

c. Next 100 to 200 units – Rs. 5.26/unit

d. above 200 units – Rs. 8.45/unit

In [26]:
```
unit = int(input("Enter Bill Units :"))
if unit <= 50:
    bill = unit*2.60
elif unit <= 100:
    bill = 130 + (unit-50)*3.25
elif unit <= 200:
    bill = 130 + 162.5 + (unit-100)*5.26
elif unit > 200:
    bill = 130 + 162.5 + 526 + (unit-200)*8.45

print(f"Bill for {unit} Units = {bill}")
```

```
Bill for 100 Units = 292.5
```

In [ ]: